



Andreas Garcia

Data & Software Engineer

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ABOUT

Engineer with a **computer science degree**, I work where data, code and clients meet. At Cleversys & **Deloitte**, I took **BI projects** from client workshops to **Tableau dashboard** presentations, and acted as technical point of contact on an ERP migration for a client with €4B in revenue. At **INRAE**, I helped **researchers** share their data on an agrivoltaics project, which turned climate change mitigation into a lasting interest.

I founded **TheMusicTree**, where I build **TheMusicTreePipelines**, **Python and Polars pipelines** ingesting **MusicBrainz** and **Wikidata** with data-quality checks. They feed **GrowTheMusicTree**, a D3.js genre map backed by **30.9M songs** in **PostgreSQL**. I run the whole stack myself, from **Ansible** provisioning to monitoring, and analyse the resulting genre graph in **Jupyter notebooks**.

SKILLS

- **Tech:** Python, SQL, PostgreSQL, Polars, DuckDB, Jupyter, D3.js, Tableau, Docker, CI/CD
- **Workflow:** Data Pipelines, Data Quality Checks, Data Visualisation, Client Workshops, Specification
- **Languages:** French (native), **English** (advanced), Spanish (advanced), German (basic)
- **Hobbies:** WWOOFing, natural building (1 year), organizer of a music festival (80+ attendees)

EXPERIENCE

TheMusicTree - Founder & Developer

July 17

Dec. 2021 – Present Paris, France

- Building **TheMusicTreePipelines**, **Python/Polars** pipelines ingesting **MusicBrainz** data and the live **Wikidata** SPARQL endpoint, with data-quality checks and a 90% coverage threshold in CI.
- Building **GrowTheMusicTree**, an interactive D3.js map of 6,000+ genres whose Django REST Framework and PostgreSQL backend loads **30.9M songs** through staged bulk imports from the pipeline outputs.
- Operating the ecosystem myself: **Ansible** and **GitHub Actions provisioning**, orchestrated pipeline runs, Prometheus and Grafana monitoring, and offsite backups to **Cloudflare R2** with a restore drill.

INRAE - Research Engineer

July 17

Oct. 2018 – Jan. 2020 Montpellier, France

- Helped **researchers** share their data through a **common vocabulary and structure** on a **cross-disciplinary agrivoltaics** project, combining agriculture and photovoltaics to **mitigate climate change**.
- Contributed to **OpenSILEX**, an open-source information system making research data **FAIR** for scientists, manipulating complex graph data in **RDF4J** and improving the underlying ontologies.
- Implemented event-handling REST services in Java with the DTO pattern, wired into the PHP 7 (Yii2) and JavaScript web layer, within a SCRUM team practising peer code reviews on **GitHub**.

D. Cleversys → Deloitte - BI/ERP Consultant

July 17

Apr. 2015 – Oct. 2017 Paris, France

- Took **BI projects** for retail, rail catering and fashion clients from conception to completion, gathering needs in client exchanges and presenting the resulting **Tableau dashboards** to them.
- Managed **data transfers** into **data warehouses** with **SSIS** and **SQL Server**, then specified and implemented **62 critical accounting** interfaces as complex **SQL scripts** for an **ERP migration project**.
- Served as **technical point of contact** for a client with annual revenue above **€4B**, leading technical-functional specification workshops and **training developers on the ERP's ETL tools**.

EDUCATION

R La Ruche - Programme Itinéraire (Entrepreneurship Program)

July 17

2025 Paris, France

- Three-month entrepreneurship program with nine workshops (marketing, strategy, legal, etc.).
- Conducted market studies and developed a business plan.

INSA Lyon - Engineering degree in Computer Science

July 17

2014 Lyon, France